

# CSE 881: Data Mining

---

## Lecture 2: Data

1

## Outline

---

- Attributes and Objects
- Types of Data
- Data Quality
- Data Preprocessing

2

# What is Data?

- Collection of data objects and their attributes
- Attribute is a property or characteristic of an object
  - Examples: eye color of a person, temperature, etc.
  - Attribute is also known as variable, field, characteristic, feature, or observation
- A collection of attributes describe an object
  - Object is also known as record, point, case, sample, entity, or instance

**Attributes**

| <i>Tid</i> | Refund | Marital Status | Taxable Income | Cheat |
|------------|--------|----------------|----------------|-------|
| 1          | Yes    | Single         | 125K           | No    |
| 2          | No     | Married        | 100K           | No    |
| 3          | No     | Single         | 70K            | No    |
| 4          | Yes    | Married        | 120K           | No    |
| 5          | No     | Divorced       | 95K            | Yes   |
| 6          | No     | Married        | 60K            | No    |
| 7          | Yes    | Divorced       | 220K           | No    |
| 8          | No     | Single         | 85K            | Yes   |
| 9          | No     | Married        | 75K            | No    |
| 10         | No     | Single         | 90K            | Yes   |

**Objects**

3

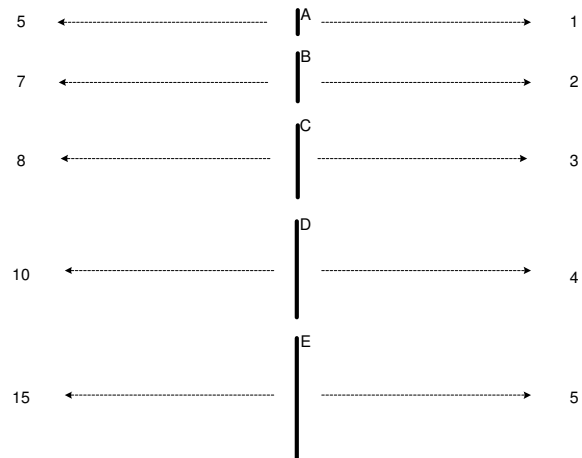
# Attribute Values

- Attribute values are numbers or symbols assigned to an attribute
- Distinction between attributes and attribute values
  - Same attribute can be mapped to different attribute values
    - ◆ Example: height can be measured in feet or meters
  - Different attributes can be mapped to the same set of values
    - ◆ Example: Attribute values for ID and age are integers
    - ◆ But properties of attribute values can be different
      - ID has no limit but age has a maximum and minimum value

4

## Measurement Scale

- The way you measure an attribute may not match the attribute's properties



5

## Types of Attributes

- There are different types of attributes
  - **Nominal**
    - ◆ Examples: ID numbers, eye color, zip codes
  - **Ordinal**
    - ◆ Examples: rankings (e.g., taste of potato chips on a scale from 1-10), grades, height in {tall, medium, short}
  - **Interval**
    - ◆ Examples: calendar dates, temperatures in Celsius or Fahrenheit.
  - **Ratio**
    - ◆ Examples: temperature in Kelvin, length, time, counts

6

## Properties of Attribute Values

- The type of an attribute depends on which of the following properties it possess:
  - Distinctness:  $= \neq$
  - Order:  $< >$
  - Addition:  $+ -$
  - Multiplication:  $* /$
  - Nominal attribute: distinctness
  - Ordinal attribute: distinctness & order
  - Interval attribute: distinctness, order & addition
  - Ratio attribute: all 4 properties

7

| Attribute Type | Description                                                                                                                                                                | Examples                                                                                  | Operations                                                         |
|----------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------|--------------------------------------------------------------------|
| Nominal        | The values of a nominal attribute are just different names, i.e., nominal attributes provide only enough information to distinguish one object from another. ( $=, \neq$ ) | zip codes, employee ID numbers, eye color, sex: { <i>male, female</i> }                   | mode, entropy, contingency correlation, $\chi^2$ test              |
| Ordinal        | The values of an ordinal attribute provide enough information to order objects. ( $<, >$ )                                                                                 | hardness of minerals, { <i>good, better, best</i> }, grades, street numbers               | median, percentiles, rank correlation, run tests, sign tests       |
| Interval       | For interval attributes, the differences between values are meaningful, i.e., a unit of measurement exists. ( $+, -$ )                                                     | calendar dates, temperature in Celsius or Fahrenheit                                      | mean, standard deviation, Pearson's correlation, $t$ and $F$ tests |
| Ratio          | For ratio variables, both differences and ratios are meaningful. ( $*, /$ )                                                                                                | temperature in Kelvin, monetary quantities, counts, age, mass, length, electrical current | geometric mean, harmonic mean, percent variation                   |

| Attribute Level | Transformation                                                                                                    | Comments                                                                                                                              |
|-----------------|-------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------|
| Nominal         | Any permutation of values                                                                                         | If all employee ID numbers were reassigned, would it make any difference?                                                             |
| Ordinal         | An order preserving change of values, i.e.,<br>$new\_value = f(old\_value)$<br>where $f$ is a monotonic function. | An attribute encompassing the notion of good, better best can be represented equally well by the values {1, 2, 3} or by {0.5, 1, 10}. |
| Interval        | $new\_value = a * old\_value + b$<br>where $a$ and $b$ are constants                                              | Thus, the Fahrenheit and Celsius temperature scales differ in terms of where their zero value is and the size of a unit (degree).     |
| Ratio           | $new\_value = a * old\_value$                                                                                     | Length can be measured in meters or feet.                                                                                             |

## Discrete and Continuous Attributes

- Discrete Attribute

- Has only a finite or countably infinite set of values
- Examples: zip codes, counts, or the set of words in a collection of documents
- Often represented as integer variables.
- Note: binary attributes are a special case of discrete attributes

- Continuous Attribute

- Has real numbers as attribute values
- Examples: temperature, height, or weight.
- Practically, real values can be measured and represented using a finite number of digits.
- Continuous attributes are typically represented as floating-point variables.

## Asymmetric Attributes

- Only presence (a non-zero attribute value) is regarded as important
- Examples:
  - Words present in documents
  - Items present in customer transactions

|            | learn | coach | play | ball | score | game | win | lost | timeout | season |
|------------|-------|-------|------|------|-------|------|-----|------|---------|--------|
| Document 1 | 3     | 0     | 5    | 0    | 2     | 6    | 0   | 2    | 0       | 2      |
| Document 2 | 0     | 7     | 0    | 2    | 1     | 0    | 0   | 3    | 0       | 0      |
| Document 3 | 0     | 1     | 0    | 0    | 1     | 2    | 2   | 0    | 3       | 0      |

11

## Types of Data Sets

- Common Types
  - Record
  - Graph
  - Ordered

12

## Record Data

- Data that consists of a collection of records, each of which consists of a fixed set of attributes

| <i>Tid</i> | Refund | Marital Status | Taxable Income | Cheat |
|------------|--------|----------------|----------------|-------|
| 1          | Yes    | Single         | 125K           | No    |
| 2          | No     | Married        | 100K           | No    |
| 3          | No     | Single         | 70K            | No    |
| 4          | Yes    | Married        | 120K           | No    |
| 5          | No     | Divorced       | 95K            | Yes   |
| 6          | No     | Married        | 60K            | No    |
| 7          | Yes    | Divorced       | 220K           | No    |
| 8          | No     | Single         | 85K            | Yes   |
| 9          | No     | Married        | 75K            | No    |
| 10         | No     | Single         | 90K            | Yes   |

13

## Data Matrix

- If data objects have the same fixed set of numeric attributes, then the data objects can be thought of as points in a multi-dimensional space, where each dimension represents a distinct attribute
- Such data set can be represented by an  $m$  by  $n$  matrix, where there are  $m$  rows, one for each object, and  $n$  columns, one for each attribute

| Projection of x Load | Projection of y load | Distance | Load | Thickness |
|----------------------|----------------------|----------|------|-----------|
| 10.23                | 5.27                 | 15.22    | 2.7  | 1.2       |
| 12.65                | 6.25                 | 16.22    | 2.2  | 1.1       |

14

## Document Data

- Each document becomes a 'term' vector,
  - each term is a component (attribute) of the vector,
  - the value of each component is the number of times the corresponding term occurs in the document.

|            | team | coach | play | ball | score | game | win | lost | timeout | season |
|------------|------|-------|------|------|-------|------|-----|------|---------|--------|
| Document 1 | 3    | 0     | 5    | 0    | 2     | 6    | 0   | 2    | 0       | 2      |
| Document 2 | 0    | 7     | 0    | 2    | 1     | 0    | 0   | 3    | 0       | 0      |
| Document 3 | 0    | 1     | 0    | 0    | 1     | 2    | 2   | 0    | 3       | 0      |

15

## Transaction Data

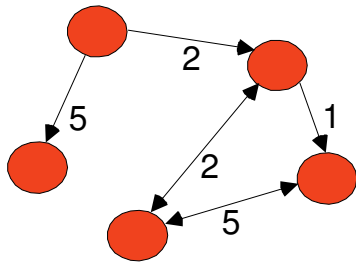
- A special type of record data, where
  - each record (transaction) involves a set of items.
  - For example, consider a grocery store. The set of products purchased by a customer during one shopping trip constitute a transaction, while the individual products that were purchased are the items.

| <i>TID</i> | <i>Items</i>              |
|------------|---------------------------|
| 1          | Bread, Coke, Milk         |
| 2          | Beer, Bread               |
| 3          | Beer, Coke, Diaper, Milk  |
| 4          | Beer, Bread, Diaper, Milk |
| 5          | Coke, Diaper, Milk        |

16

## Graph Data

- Examples: Generic graph and HTML Links

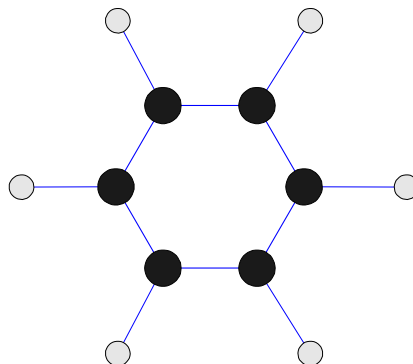


```
<a href="papers/papers.html#bbbb">  
Data Mining </a>  
<li>  
<a href="papers/papers.html#aaaa">  
Graph Partitioning </a>  
<li>  
<a href="papers/papers.html#aaaa">  
Parallel Solution of Sparse Linear System of Equations </a>  
<li>  
<a href="papers/papers.html#ffff">  
N-Body Computation and Dense Linear System Solvers
```

17

## Chemical Data

- Benzene Molecule:  $C_6H_6$

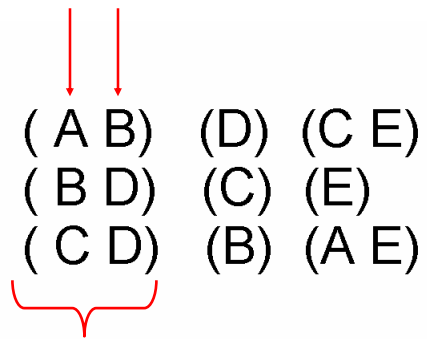


18

## Ordered Data

- Sequences of transactions

Items/Events



19

## Ordered Data

- Genomic sequence data

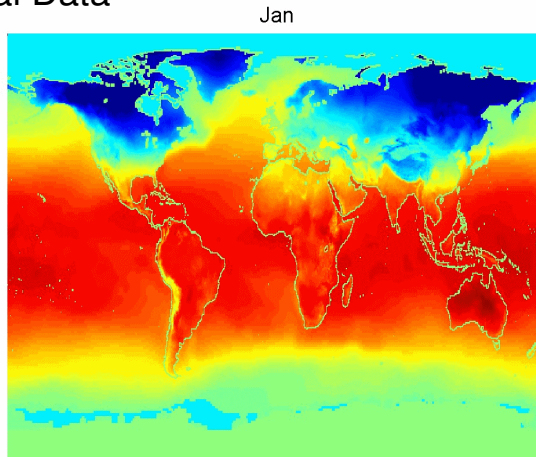
```
GGTTCCGCCTTCAGCCCCGCGCC
CGCAGGGCCCGCCCCGCGCCGTC
GAGAAGGGCCCGCCTGGCGGGCG
GGGGGAGGCGGGGCCGCCGAGC
CCAACCGAGTCCGACCAGGTGCC
CCCTCTGCTCGGCCTAGACCTGA
GCTCATTAGGCGGCAGCGGACAG
GCCAAGTAGAACACGCGAAGCGC
TGGGCTGCCTGCTGCGACCAGGG
```

20

## Ordered Data

- Spatio-Temporal Data

Average Monthly  
Temperature of  
land and ocean



21

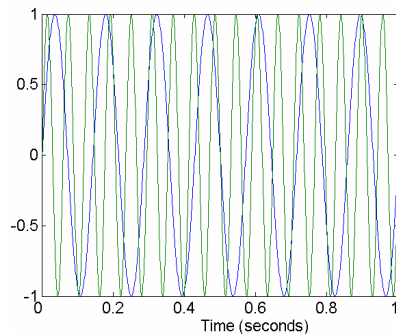
## Data Quality Issues

- Noise
- Outliers
- Missing values
- Duplicate data

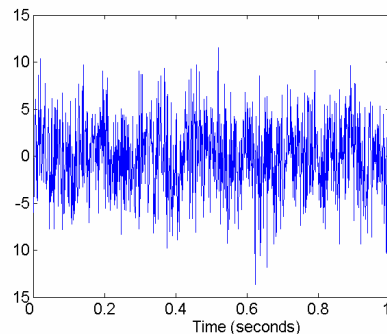
22

## Noise

- Noise refers to modification of original values
  - Examples: distortion of a person's voice when talking on a poor phone and “snow” on television screen



Two Sine Waves

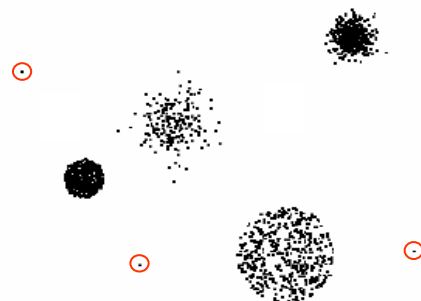


Two Sine Waves + Noise

23

## Outliers

- Outliers are data objects with characteristics that are considerably different than most of the other data objects in the data set



24

## Missing Values

---

- Reasons for missing values
  - Information is not collected (e.g., people decline to give their age and weight)
  - Attributes may not be applicable to all cases (e.g., annual income is not applicable to children)
- Handling missing values
  - Eliminate Data Objects
  - Estimate Missing Values
  - Ignore the Missing Value During Analysis
  - Replace with all possible values (weighted by their probabilities)

25

## Duplicate Data

---

- Data set may include data objects that are duplicates, or almost duplicates of one another
  - Major issue when merging data from heterogeneous sources
- Examples:
  - Same person with multiple email addresses
- Deduplication
  - Process of dealing with duplicate data issues

26

## Data Preprocessing

---

- Aggregation
- Sampling
- Dimensionality Reduction
- Feature subset selection
- Feature creation
- Discretization and Binarization
- Attribute Transformation

27

## Aggregation

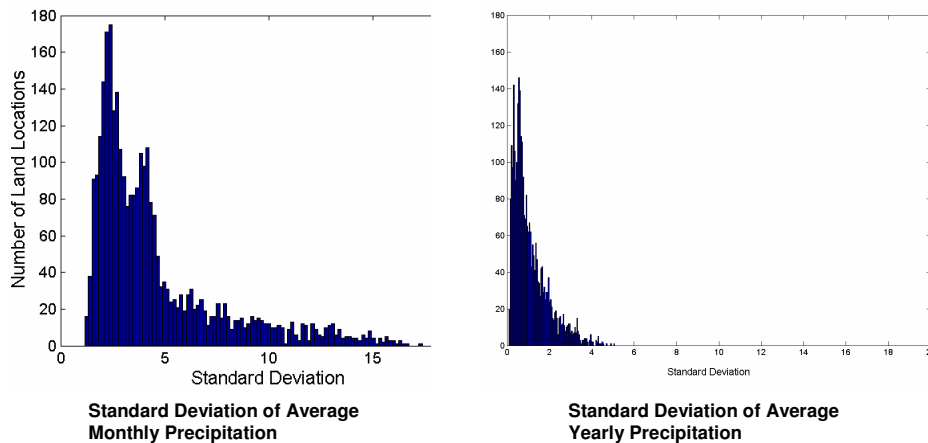
---

- Combining two or more attributes (or objects) into a single attribute (or object)
- Purpose
  - Data reduction
    - ◆ reduce the number of attributes or objects
  - Change of scale
    - ◆ cities aggregated into regions, states, countries, etc
  - More “stable” data
    - ◆ aggregated data tends to have less variability

28

## Aggregation

### Variation of Precipitation in Australia



29

## Sampling

- Sampling is the main technique employed for data selection
  - It is often used for both the preliminary investigation of the data and the final data analysis
- Statisticians sample because obtaining the entire set of data of interest is too expensive or time consuming
- Sampling is used in data mining because it is too expensive or time consuming to process all the data

30

## Sampling ...

---

- The key principle for effective sampling is the following:
  - using a sample will work almost as well as using the entire data sets, if the sample is representative
  - A sample is representative if it has approximately the same property (of interest) as the original set of data

31

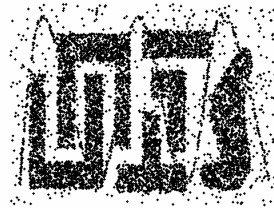
## Types of Sampling

---

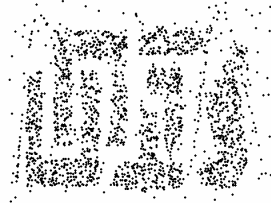
- Simple Random Sampling
  - There is an equal probability of selecting any particular item
- Sampling without replacement
  - As each item is selected, it is removed from the population
- Sampling with replacement
  - Objects are not removed from the population as they are selected for the sample.
    - ◆ In sampling with replacement, the same object can be picked up more than once
- Stratified sampling
  - Split the data into several partitions; then draw random samples from each partition

32

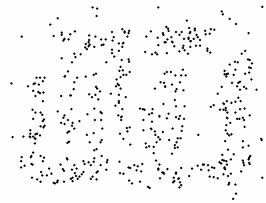
## Sample Size



8000 points



2000 Points

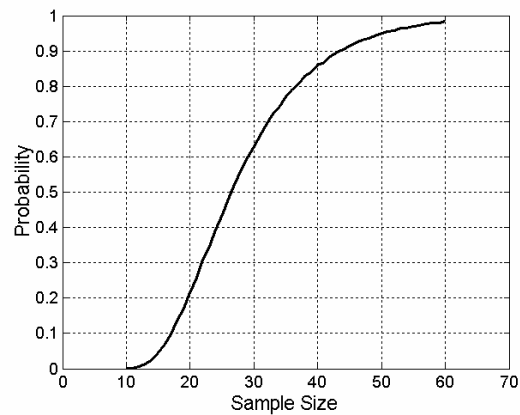


500 Points

33

## Sample Size

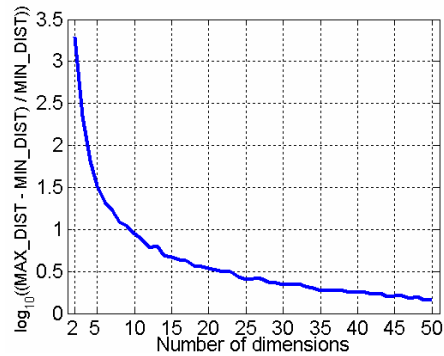
- What sample size is necessary to get at least one object from each of 10 groups.



34

## Curse of Dimensionality

- When dimensionality increases, data becomes increasingly sparse in the space that it occupies
- Definitions of density and distance between points, which is critical for clustering and outlier detection, become less meaningful



- Randomly generate 500 points
- Compute difference between max and min distance between any pair of points

35

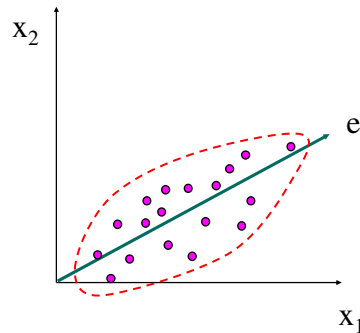
## Dimensionality Reduction

- Purpose:
  - Avoid curse of dimensionality
  - Reduce amount of time and memory required by data mining algorithms
  - Allow data to be more easily visualized
  - May help to eliminate irrelevant features or reduce noise
- Techniques
  - Principle Component Analysis
  - Singular Value Decomposition
  - Others: supervised and non-linear techniques

36

## Dimensionality Reduction: PCA

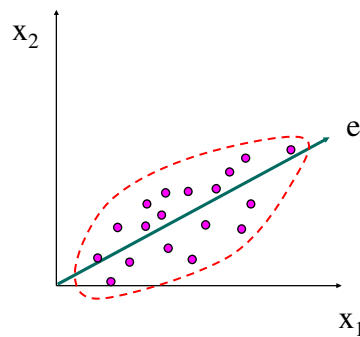
- Goal is to find a projection that captures the largest amount of variation in data



37

## Dimensionality Reduction: PCA

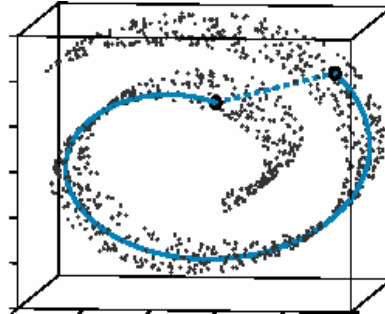
- Find the eigenvectors of the covariance matrix
- The eigenvectors define the new space



38

## Dimensionality Reduction: ISOMAP

By: Tenenbaum, de Silva,  
Langford (2000)



- Construct a neighbourhood graph
- For each pair of points in the graph, compute the shortest path distances – geodesic distances

39

## Feature Subset Selection

- Another way to reduce dimensionality of data
- Redundant features
  - duplicate much or all of the information contained in one or more other attributes
  - Example: purchase price of a product and the amount of sales tax paid
- Irrelevant features
  - contain no information that is useful for the data mining task at hand
  - Example: students' ID is often irrelevant to the task of predicting students' GPA

40

## Feature Subset Selection

---

- Techniques for selecting best subset:
  - Embedded approaches:
    - ◆ Feature selection occurs naturally as part of the data mining algorithm
  - Filter approaches:
    - ◆ Features are selected before data mining algorithm is run
  - Wrapper approaches:
    - ◆ Use the data mining algorithm as a black box to find best subset of attributes

41

## Feature Creation

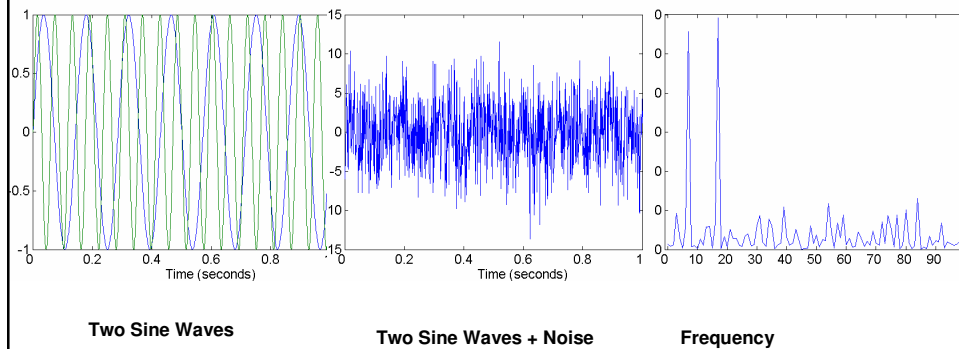
---

- Create new attributes that can capture the important information in a data set much more efficiently than the original attributes
- Three general methodologies:
  - Feature Extraction
    - ◆ domain-specific
  - Mapping Data to New Space
  - Feature Construction
    - ◆ combining features

42

## Mapping Data to a New Space

- Fourier transform
- Wavelet transform



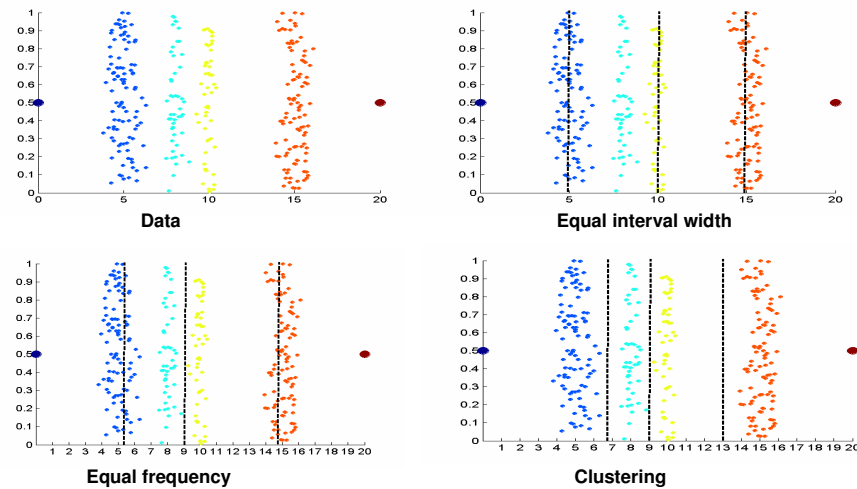
43

## Discretization

- Applied to continuous attributes
- Typically used in classification and association analysis

44

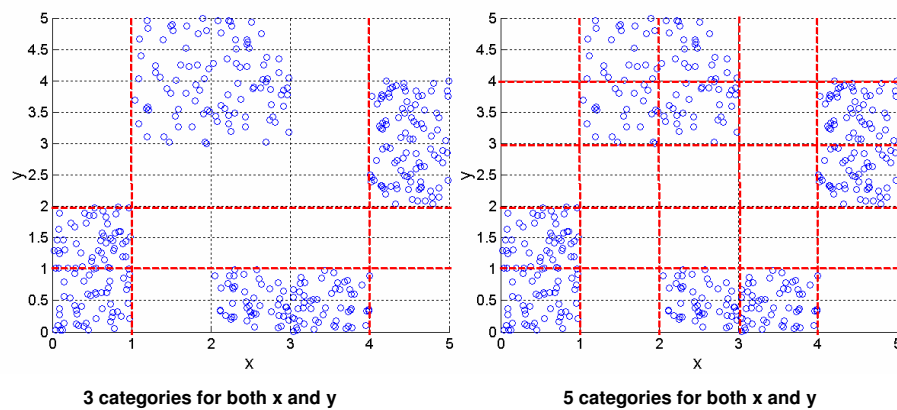
# Unsupervised Discretization



45

# Supervised Discretization

- Entropy based approach – discretization with class labels



46

## Attribute Transformation

- A function that maps the entire set of values of a given attribute to a new set of replacement values such that each old value can be identified with one of the new values
  - Simple functions:  $x^k$ ,  $\log(x)$ ,  $e^x$ ,  $|x|$
  - Standardization and Normalization

